Classification of elements and Periodic properties

Periodic table is arrangement of chemical elements in a tabular form on basis of their properties which facilitates the systematic study of properties of the elements.

History of Periodic Table:

Dobereiner's Traid:- Tried - set of three elements He arranged the elements in set of three in such a way that atomic wt of middle element is arithmatic mean of other two other two element.

eg: Li = 7, Na = 23, K = 39 ⇒ $\frac{7+39}{2} = \frac{46}{2} = 23$

Cl	35.5	$\frac{127+35.5}{2}$
Br	80	
I	127	$= 81.25$

* All element do not follow Dobereiner's Traid

eg: F 19, Cl 35.5, Br 80 $\quad \frac{19+80}{2} = \underline{\underline{49.5}}$

* It was only applicable for group.

Newland Law of octaves: Octave - eight (8)

(Sa)[1] Re[2] Ga[3] Ma[4] Pa[5] Dha[6] Ni[7] (Sa)[8]

→ First Scientist who related Properties of element with atomic mass.

→ According to Newland when element are arranged in increasing order of atomic wt then every 8th element will have same Property like 1st element.

[Li]	Be	B	C	N	O	F	* Valid only upto Calcium
[Na]	Mg	Al	Si	P	S	Cl	Failed after discovery of Inert gas.
K	Ca						

Mendeleev's periodic Law: "The physical and chemical properties of elements are periodic functions of their atomic masses."

Drawbacks:

- ✓ No place for isotopes
- ✓ In few places, element having more atomic mass was placed before element having lower atomic mass.
- ✓ No unique place for H.

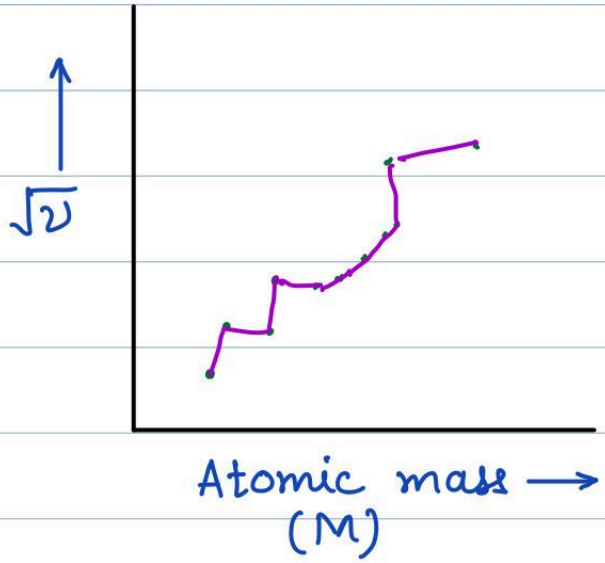
But Mendeleev's periodic table was challenged by Moseley's experiment.

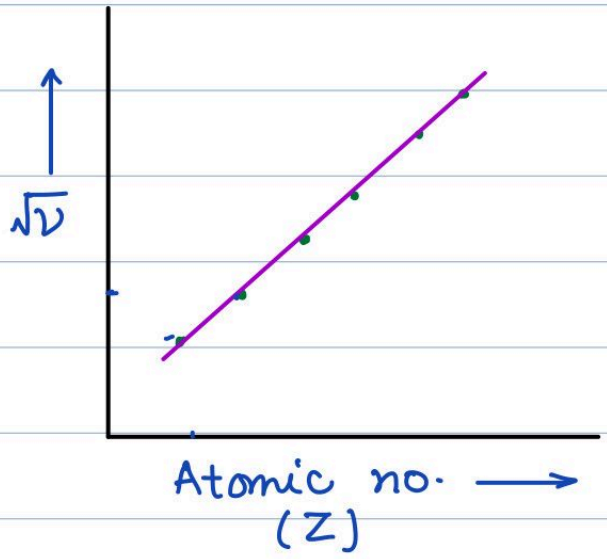
Moseley's experiment:

(Basis of modern periodic table)

He bombarded high speed (energetic) electrons on different heavy metal surfaces and obtained X-rays.

He observed a systematic mathematical relation between the frequency of emitted X-rays and atomic no. of metals used.





$$\sqrt{\nu} = a(Z-b)$$ Moseley's eqⁿ

- ν = frequency of emitted X-rays
- Z = atomic no. element
- a, b = constants

$$\sqrt{\nu} \propto Z$$

Modern Periodic Law:

The physical and chemical properties of elements are periodic functions of their atomic numbers.

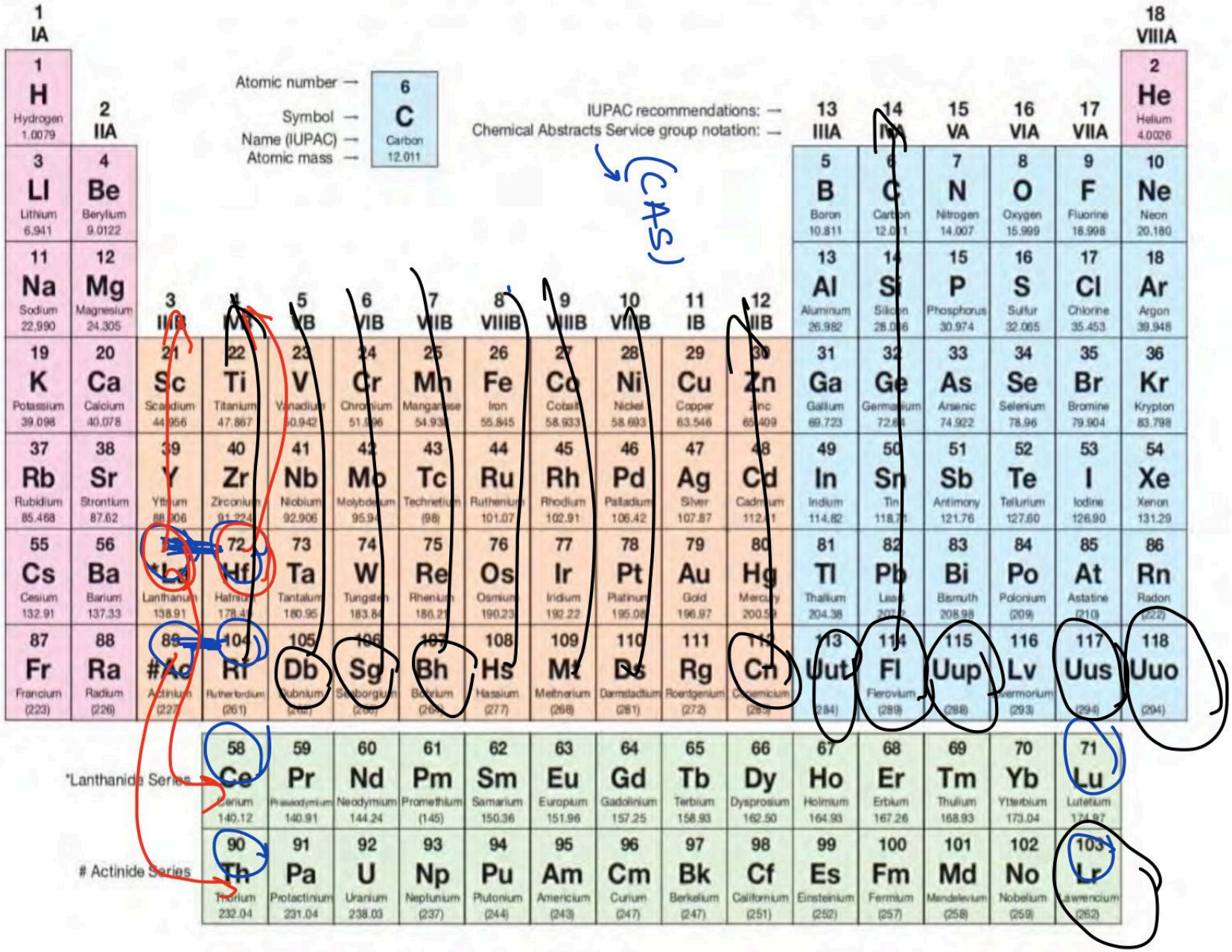
i.e. if elements are arranged in order of increasing atomic numbers, the elements with similar properties are repeated after certain regular intervals.

Modern periodic law is essentially the consequence of periodic variation in electronic configeration.

Long Form of modern periodic table:

- Total no. of known elements = 118
- Horizontal rows of periodic table are called "Periods" [Total no. of periods = 7]
- vertical columns of periodic table are called "Groups" (or) "families" [Total no. of Groups = 18]
- Total no. of blocks = 4 (s, p, d, f)

Periodic Table of the Elements



No. of elements in periods:

Period No.	No. of elements
1	2
2	8
3	8
4	18
5	18
6	32
7	32

Blockwise brief introduction of periodic table:

[A] **s-block:** (all metals)

- ✓ contains Group-1 & Group-2 metals
- ✓ General configuration: [noble gas] $ns^{1 \text{ or } 2}$
- Group-1 metals: known as alkali metals

  ∵ they form alkaline hydroxide on rxn with H_2O.

- Group-2 metals: known as alkaline earth metals (except Be)

  ∵ They form alkaline oxides and hydroxides and their oxides are abundant in earth's crust.

BeO (or) $Be(OH)_2$: Amphoteric

So, Be is not included in alkaline earth metal.

[B] p-Block

- Contains metals, non-metals, metalloids & noble gases.
- Group no. 13 to Group no. 18 → p-block
- General configuration = [noble gas] ns^2 $np^{1\text{ to }6}$

pnictogens (Gr-15 element)	chalcogens (Gr-16 elements)	halogens (Gr-17 elements)
N, P, As, Sb, Bi	(O, S, Se, Te, Po)	(F, Cl, Br, I, At)
→ poisonous	→ ore forming	→ salt forming

Noble gases / inert gases:

- Group-18 elements (He, Ne, Ar, Kr, Xe ---)
- Least reactive

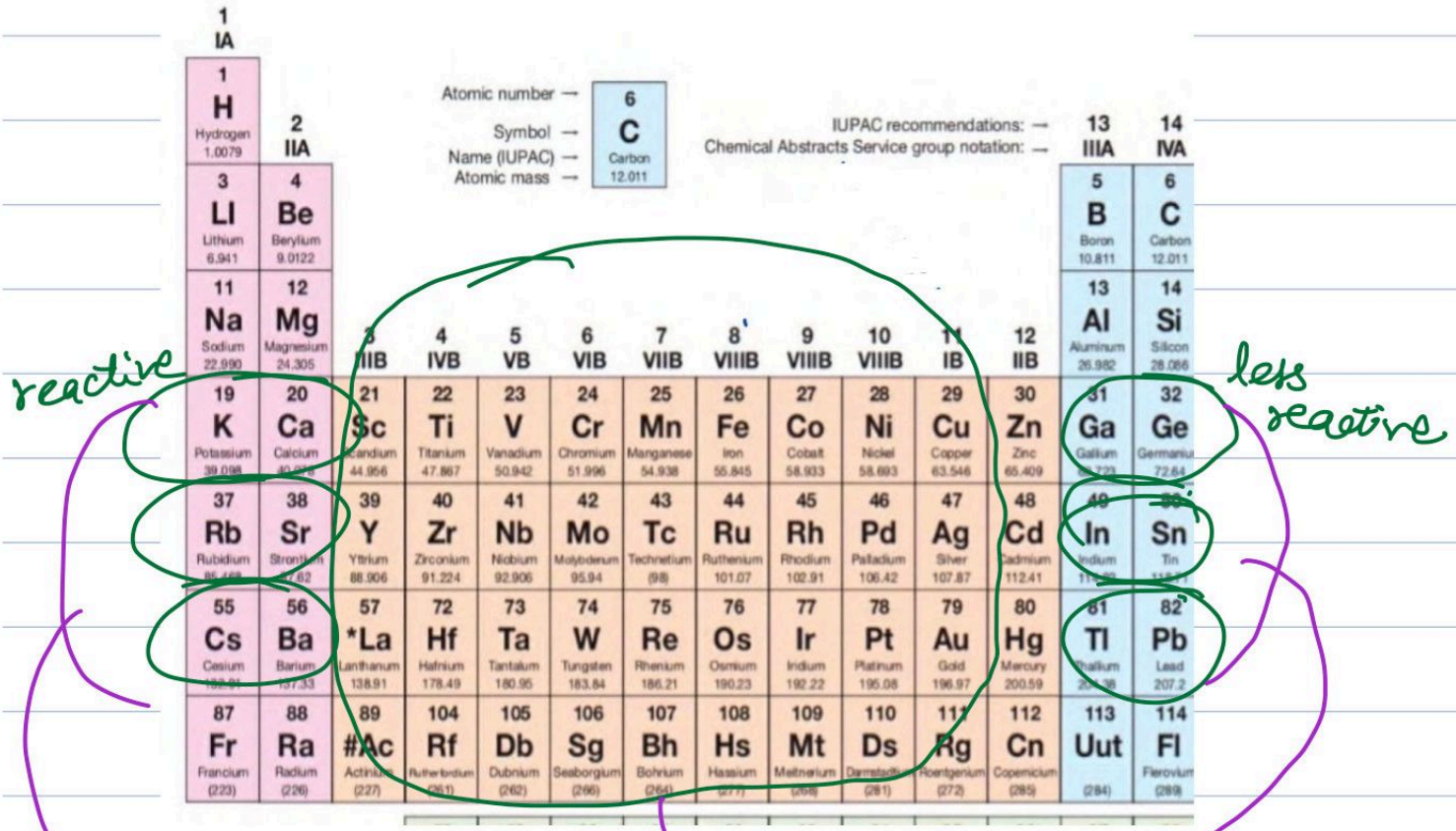
[C]. d-Block:

- ✓ all metals
- ✓ contains → Group-3 to Gr-12
- ✓ General configuration: [noble gas] $ns^{1\text{ to }2}$ $(n-1)d^{1\text{ to }10}$

- All d-block metals except Group no. 12 (Zn, Cd, Hg) are called transition elements (or)

Transition metals.

Transition elements form a bridge between reactive metals of s-block and less reactive metals of p-block (i.e. Group-13).



d¹ to d⁹ configuration in atom or ions

either d^0 or d^{10} configuration in atom or ion

Zn, Cd, Hg $\xrightarrow{\text{either atom or ion}}$ all have d^{10}

→ not included in transition element.

- In atom or ion of transition elements, they have partially filled d-subshell.

Q Cu has configuration $[Ar]\ 4s^1\ 3d^{10}$. Still it is included in transition elements. How?

Sol: Cu^{2+} (ion) : $[Ar]\ 4s^0\ 3d^9$

Zn, Cd, Hg ⟹ Zn: $[Ar]\ 4s^2\ 3d^{10}$ → fully filled d
Zn^{2+}: $[Ar]\ 4s^0\ 3d^{10}$ → fully filled d

[D] f-block:

- ✓ all metals
- ✓ Group-3
  - at. no. 58 to 71: Lanthanoids (or) Lanthanides
  - at. no. 90 to 103: Actinoids (or) Actinides
- ✓ General configuration: $[\text{noble gas}]\ ns^2 (n-1)d^{0\text{ to }1} (n-2)f^{1\text{ to }14}$

- They are called inner transition elements. (at. no. in b/w transition elements)

[Lanthanoids: 14 elements lying after lanthanum
[Actinoids: 14 elements lying after actinium

Transurenic / transurenium elements:

- Elements after at. no. 92 (Uranium)
- They are man-made (synthetic) not known in nature.

Representative elements / Main group element

Representative elements / Main group elements:

- Elements of s-block and p-block are collectively known as main group elements.

Finding group no., Block, & period no. for an element (whose atomic no. is given) in Periodic table:

Method '1' (for school exams):

Step: 1. Write electronic configuration

Step: 2. max. principal quantum no. = period no.

Step: 3. Subshell of last e^- = Block

Step: 4. To find group no, we use following table:

Block	Group no.
s	= no. of $\bar{e}$s in outermost s
p	= no. of $\bar{e}$s in outermost p + 12
d	= no. of $\bar{e}$s in outermost $_{(ns)}$ s + no. of $\bar{e}$ in inner $_{(n-1)}$ d
f	= 3

ex: atomic no. 15

$= 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

- Period = 3rd
- Block = p
- Group no. = (3+12) = 15

ex at. no. 24

$1s^2\ 2s^2 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

- Period = 4th ($\rightarrow$ ns)
- Block = d ($\rightarrow$ (n-1) d)
- Group no. = (1+5) = 6

---

atomic no. 12

$1s^2\ 2s^2\ 2p^6\ 3s^2$

- Period = 3rd
- Block = s
- Group no. = 2

atomic no. 62

$[^{54}Xe]\ 6s^2\ 4f^6$

- Period = 6th
- Block = f
- Group no. = 3

---

* It is failed in some cases:

Pd(46): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 4d^{10}$ (actual configuration)

Period acc. to configuration: 4th

But actually its period is: 5th

Method '2'.

- s-block $\Rightarrow$ Gr 1 & 2
- p- " $\Rightarrow$ Gr 13 to Gr 18
- d- " $\Rightarrow$ Gr 3 to Gr 12
- f-block $\Rightarrow$ at. no. 58 to 71, at. no. 90 to 103

Gr: no. 18	Period
He(2)	1st
Ne(10)	2nd
Ar(18)	3rd
Kr(36)	4th
Xe(54)	5th
Rn(86)	6th

ex at. no. 15

Period: 3rd
Gr: no. 15
p-block

at. no. 53

Period : 5th
Gr: no. : 17
p-block

He
Ne
Ar
Kr
Xe
Rh
Og

18 - (86-81)
⇒ 18 - 5 = 13

Nh ⇒ 18-(118-113)
= 18-(5)
⇒ 13

(Cs) ⇒ 18-[118-55)
= 32-

this method is applicable for 4th 5th and 6th and 7th period only

(*) Find the period with the help of Noble gas

atomic Number = 53

period = 5

Step-2
shift in the observed period to find the GpNo

Gp- 1

(17) 18 - Memorise this Noble gas
53 54

group NO = (17)

Example!-

(*) atomic Number - 33

(*) Gp 18 period

He (2)	1
Ne (10)	2
Ar (18)	3
Kr (36)	4
Xe (54)	5
Rn (86)	6

period - 4th

(*) Gp Number :- (15)

(15) 16 17 18

33 34 35 36

(*) Example - 24

(1) Find period Number → period 4th

(2) 10 11 12 13 14 15 16 (17) 18

(*) 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

- atomic no. from 103 to 118:

[Group. no. = atomic No. − 100]

Prediction of atomic no. of a given location of element in periodic table:

ex (i) [Period 2nd, Group. no. 16]

→ atomic No. = 8

$16 = 18 - (10 - x)$

$16 = 18 - 10 + x$

$16 = 8 + x$

$x = 8$

ex (ii) [Period: 4th, Group: no. 5]

→ at. no = 23

Gr: 18	Period ↓
He (2)	1st
Ne (10)	2nd
Ar (18)	3rd
Kr (36)	4th
Xe (54)	5th
Rn (86)	6th

Prediction of atomic no. of elements lying above and below for an element of given atomic no:

ex (i)

at. no. 17

above element: at. no. 9

below element: at no. 35

Gr 17	Gr: 18	Period ↓
	He (2)	1st
←	Ne (10)	2nd
◎	Ar (18)	3rd
←	Kr (36)	4th
	Xe (54)	5th
	Rn (86)	6th

IUPAC naming of elements having atomic no. more than 100 (naming of superheavy elements):

- ✓ It is a temporary naming of elements.
- ✓ It is based on digits of atomic numbers of elements.

Digit	Root	abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	~~tetra~~ quad	q
5	pent	p
6	hex	h
7	~~hept~~ sept	s
8	oct	o
9	enn	e

abbreviation

abbreviation

- ✓ The common suffix is : "ium"
- ✓ Add the roots in sequence of digits shown and then add common suffix for naming.
- ✓ i + i = i

ex: at. no. 107

Un nil sept

Name: Unnilseptium

Symbol: Uns

ex: at. no. 113

un un tri

Name: Ununtrium

↳ Ununtrium

Q Find group no. of element Unq

Sol: at. no. = 104

Group. no. = 104 − 100 = 4

Q. what can be group. no. of a hypothetical element whose atomic no. is 120?

Sol: Group − 2 (s-block)

↳ alkaline earth metal

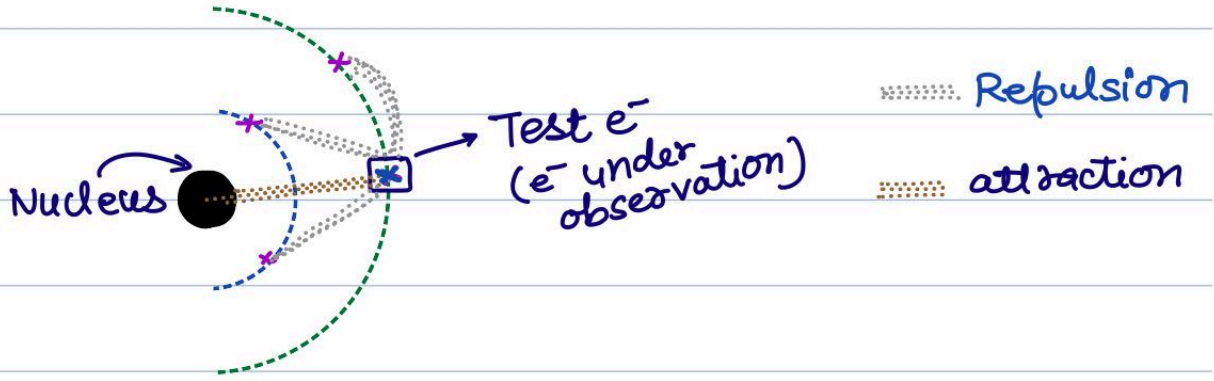
Some important effects:

[A] Penetration effect:

- ✓ Penetration describes the degree of closeness of an orbital w.r.t. the nucleus.
- ✓ For a given shell, penetration follows the order: $s > p > d > f$

[B]. Screening effect (Shielding effect):

Imp:



- Due to presence of inner electrons or other electrons present in same shell, the outer e⁻ does not experiance the full positive charge of the nucleus. (Full positive charge of nucleus = Ze)
- The effect is lowered due to the partial screening of positive charge of the nucleus by inner electrons.
- This is called shielding of outer electron from nucleus by the inner electrons or other electrons present in the same shell.

Element	configuration	Screening to outermost e⁻ by
Li(3)	$1s^2\ 2s^1$	Two ēs of 1s
Be(4)	$1s^2\ 2s^2$	Two ēs of 1s + one left ē of 2s
S(16)	$1s^2 2s^2 2p^6\ 3s^2 3p^4$	Two ēs of 1s + Two ēs of 2s + six ēs of 2p + Two ēs of 3s + three left ēs of 3p

* The order of shielding effect exerted by electrons of a given shell on a test e^- is given as: $ns > np > nd > nf$ (n = shell no.)

- within a shell, s-orbital electrons spends more time close to the nucleus in comparison to p-orbital electrons which spends more time in the vicinity of nucleus as compared to d-orbital electrons.

[C]. Effective nuclear charge ($Z_{effective}$):

The net positive charge (i.e. actual charge that electron feels) of nucleus experienced by an electron in an atom is called effective nuclear charge.

$$Z_{effective} = Z - \sigma$$

(Z: atomic no.; σ: screening const (or) shielding const)

For any single e^- species,

$\sigma = 0$ (No other e^- to shield)

$Z_{effective} = Z$

***

Across the period, effective nuclear charge increases from left to right.

2nd period:

Li	Be	B	C	N	O	F	Ne
$1s^2 2s^1$	$1s^2 2s^2$	$1s^2 2s^2 2p^1$	$1s^2 2s^2 2p^2$	$1s^2 2s^2 2p^3$	$1s^2 2s^2 2p^4$	$1s^2 2s^2 2p^5$	$1s^2 2s^2 2p^6$

→ inc. proton by 1 unit (Z inc. by 1 unit)
inc. electron by 1 unit (σ inc. but by less than 1 unit)

✓ Z inc. more rapidly than σ
hence, Z_{eff} inc.

Periodic properties:

(A) Atomic radius
(B) Ionic radius
(C) Ionisation energy
(D) Electron affinity
(E) Electronegativity

[1]. Atomic radius:

In an atom, the distance between nucleus and outermost electron cloud is atomic radius.

- Exact atomic radius of an atom can not be obtained as:

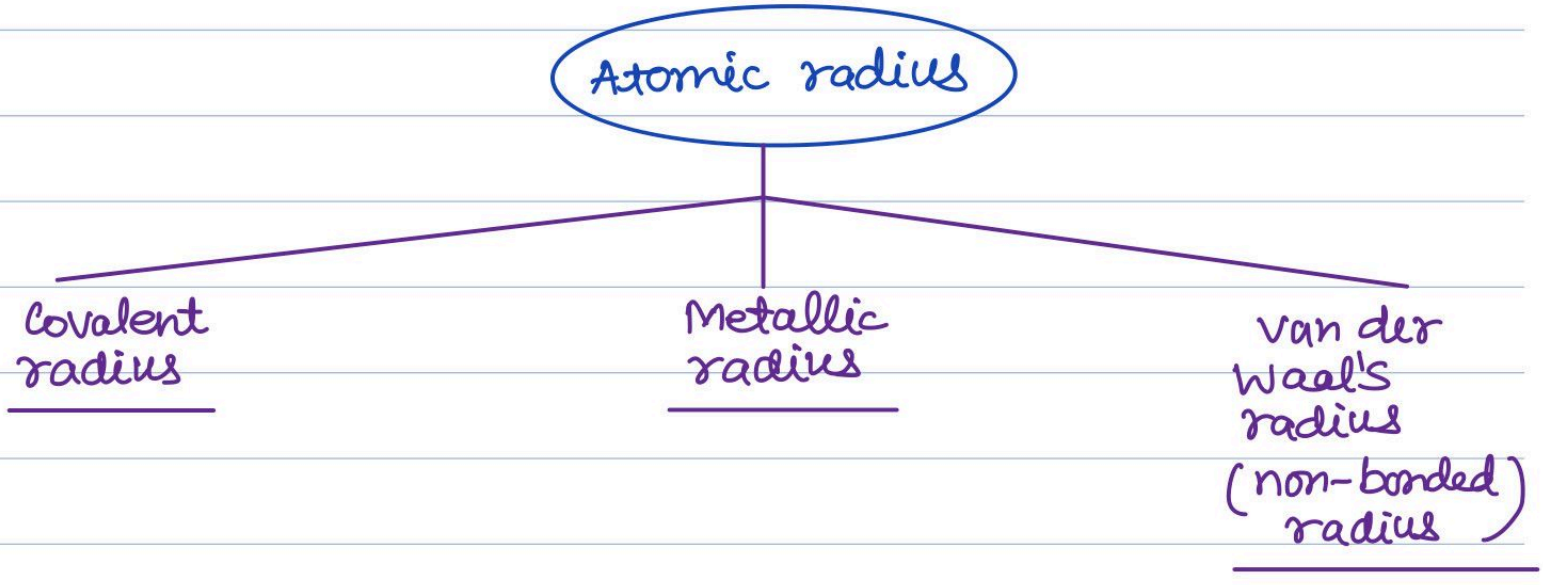
(i) size of atoms are too small (~ 1.2 Å)

(ii) there is no sharp boundary of electron cloud.

(iii) Atoms are mostly found in bonded form.

(iv) surroundings affects size

- size of atoms are studied in their respective bonded states.



Covalent radius:

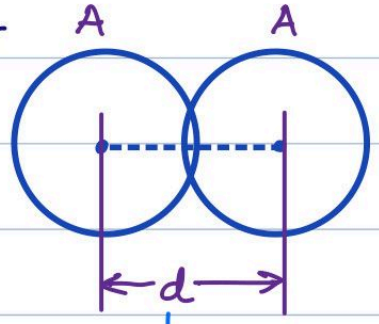
(SBCR: Single bond covalent radius)

Half of the internuclear distance in a single covalently bonded homonuclear molecule is covalent radius.

Internuclear distance: distance b/w two nucleus of bonded atoms.

Homonuclear molecule: Molecule made of same element

Homonuclear Molecule



bond length (internuclear distance)

$$(r_c) = \frac{d}{2}$$

covalent radius of atom 'A'

ex: Bond length in Cl_2 molecule = 198 pm

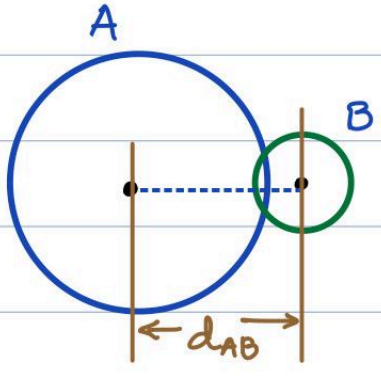
covalent radius of Cl atom = $\frac{198}{2}$ = 99 pm

✓ On increasing bond multiplicity, internuclear distance decreases.

ex:

C–C	C=C	C≡C
d = 1.54 Å	1.33 Å	1.20 Å

Heteronuclear molecule:



Stevenson-Schomaker's Equat

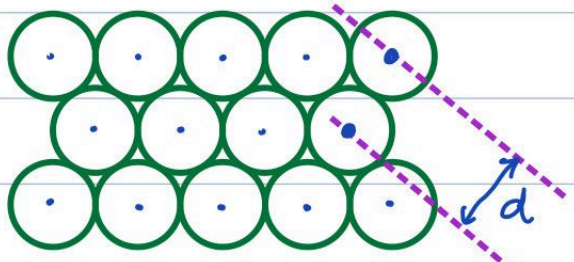
✓ Stevenson-Schomaker's eqⁿ:

$$d_{A-B} = (r_c)_A + (r_c)_B - 0.09\,|\chi_A - \chi_B|$$

- d_{A-B} = A–B bond length in Å
- $(r_c)_A$ = covalent radius of A in Å
- $(r_c)_B$ = covalent radius of B in Å
- χ_A = Electronegativity of A
- χ_B = electronegativity of B

Metallic radius:

Half of the internuclear distance of two closest atoms in a metallic crystal is called metallic radius.



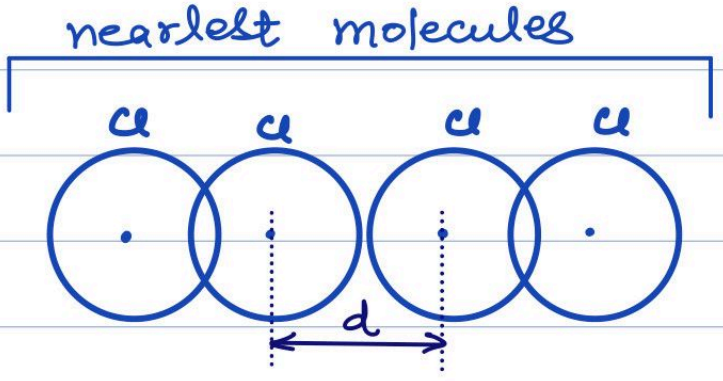
(Metallic lattice)

$$r_m = \frac{d}{2}$$

van der Waal's radius:

In solid state, half of the internuclear distance of two closest atoms belonging to the nearest molecules is called van der Waal's radius.

ex: ✓



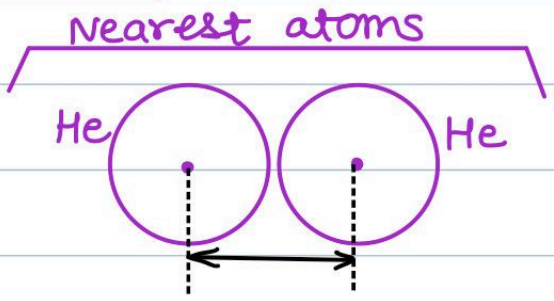
Cl_2 (solid state)

$$(r_v)_{Cl} = \frac{d}{2}$$

↳ van der Waal's radius of Cl atom

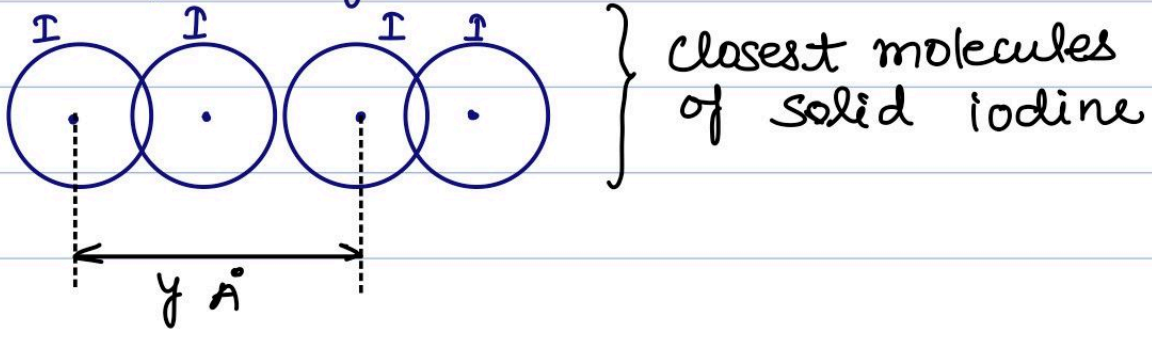
✓ For noble gases,

Solid He:

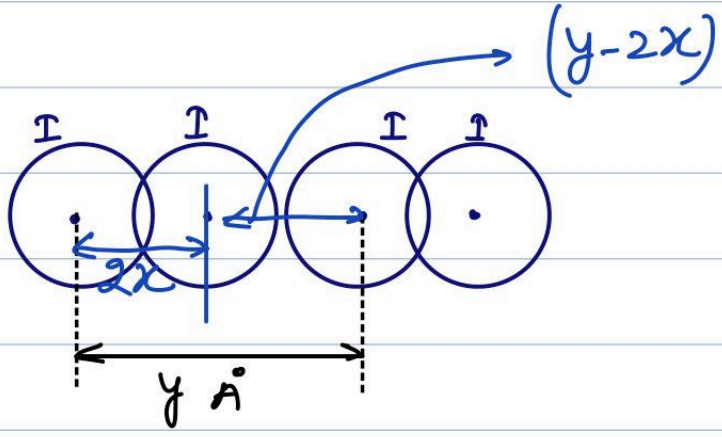


$$(r_v)_{He} = \frac{d}{2}$$

Q. covalent radius of iodine is x Å. find van der waal's radius of iodine considering the following diagram:



solution:



$$(r_v) = \frac{(y-2x)}{2}$$

⁂

If a particular element can show all these three types of radius, then: $r_v > r_m > r_c$

Factors affecting atomic radius:

(i) **Number of shells:** Generally, greater the no. of shells, more will be the atomic radius.

ex: Li : $1s^2\ 2s^1$: 2 shells

Na : $1s^2\ 2s^2\ 2p^6\ 3s^1$: 3 shells

Size of Na is more than that of Li.

(ii) **Effective nuclear charge:**

Generally, more $Z_{effective}$ → smaller size

ex:

B : $1s^2 2s^2\ 2p^1$ | C : $1s^2\ 2s^2\ 2p^2$ | N : $1s^2\ 2s^2\ 2p^3$

→

✓ Inc. $Z_{effective}$

✓ dec. size

(iii) **Bond multiplicity:**

For the same bond, Greater the bond multiplicity, smaller the atomic size.

ex: C–C C=C C≡C

→

✓ Inc. bond multiplicity

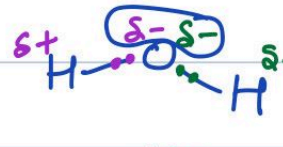
✓ dec. size of C

(iv) Oxidation number :

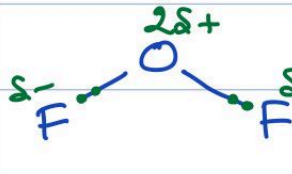
Real or hypothetical charge present on atom in its compounds is oxidation no.



Electronegativity order : F > O > N ≈ Cl > C ≈ H

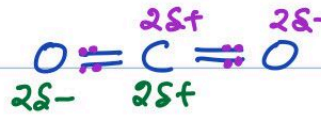


ox. no. of O = −2
ox. no. of each H = +1



ox. no. of O = +2
ox. no. of each F = −1

CO_2 :



{ ox. no. of C = +4
ox. no. of each O = −2

For the same element,

{ more +ve ox. no. ⟶ smaller size
more −ve ox. no. ⟶ larger size

$H_2\overset{-2}{O}$, $\overset{+2}{O}F_2$

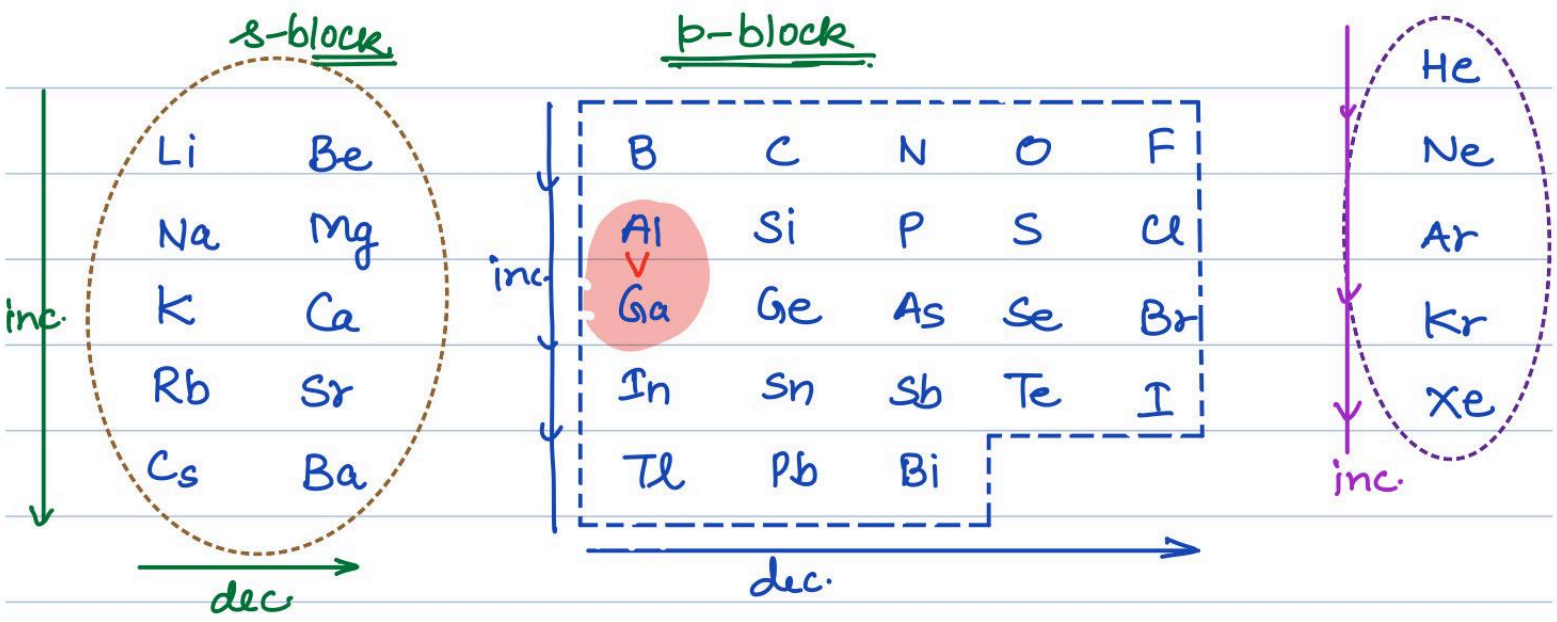


size of oxygen of water is larger than oxygen of OF_2.

General variation of atomic size in periodic table:

In a period, from left to right → size dec. (∵ Z_{eff} inc.)

In a group size inc. from top to bottom (∵ no. of shells inc.) ↓



Size of Al and Ga: (v.v. imp).

Atomic radius of Al = 1.43 Å
Atomic radius of Ga = 1.35 Å
} Don't try to learn values.

The size of Ga is slightly smaller than that of Al (If we show leniency in comparison, we can say that size of Ga is nearly equal to that of Al).

Al (13): $[Ne]\ 3s^2 3p^1$

Ga (31): $[Ar]\ 4s^2 3d^{10} 4p^1$

↓ provides poor shielding to outermost e^-

⇓

size dec. ← inc. Z_{eff}

"Transition contraction"

d-block:

size dec. →

size almost const → inc

Gr 3	Gr 4	Gr 5	Gr 6	Gr 7	Gr 8	Gr 9	Gr 10	Gr 11	Gr 12
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
La (f^0)	Hf(72) $[^{54}Xe]6s^2 4f^{14} 5d^2$	Ta	W	Re	Os	Ir	Pt	Au	Hg

explanation of variation in period:

Sc ——— Cr	inc. in $Z_{effective}$ is dominating
Cr ——— Cu	inc. in Z is compensated by inc. in $e^- - e^-$ repulsion
Cu ——— Zn	inc. in $e^- - e^-$ repulsion is dominating

explanation of variation of size in Groups:

Sc – Zn : 3d-series
Y – Cd : 4d-series
La – Hg : 5d-series

Lanthanoid contraction ⇒

From Gr-4 to Gr:12

5d-series element ⇒ $4f^{14}$ electrons (very poor shielding)

⇒ inc. $Z_{eff.}$ ⇒ size contraction

within a group, size of 5d-series element is nearly equal to that 4d-series element from Gr-4 to Gr-12.

(II). Ionic radius:

[A] Cation radius:

Cations are formed by loss of one or more electrons in atoms.
(i.e. oxidation of atoms)

$$Li \longrightarrow Li^+ + e^-$$

	Li	Li^+
No. of e^-s:	3	2
No. of protons: (atomic no.)	3	3

In Li & Li^+, Z is same & σ - different

$Z_{effective}$ is more for Li^+ than for Li.

Size of a cation is always smaller than its parent atom because cation has more $Z_{effective}$ as compared to parent atom.

[B] Anion radius:

Anions are formed by gain of one or more electrons in atom. (i.e. Reduction of atom)

$$F + e^- \longrightarrow F^-$$

	F	F^-
No. of e^-s:	9	10
No of protons: (atomic no Z)	9	9

✓ $Z_{effective}$ more for F atom than for F^- ion.

size of an anion is always larger than its parent atom because anion has lesser Zeffective as compared to parent atom.

✓ comparisons of size:

(i) $O^{2-} > O^{-} > O > O^{+} > O^{2+}$

(ii) $I^{-} > I > I^{+}$

(iii) $Fe > Fe^{+2} > Fe^{+3}$

(iv) $H^{-} > H > H^{+}$ → (proton)

comparison of size in isoelectronic species:

(i):

	O^{2-}	F^{-}	Na^{+}	Mg^{2+}
No. of $\bar{e}$s:	10	10	10	10
No. of protons:	8	9	11	12

Zeffective: $Mg^{2+} > Na^{+} > F^{-} > O^{2-}$

Size: $O^{2-} > F^{-} > Na^{+} > Mg^{2+}$

(ii).

	K^{+}	S^{2-}	Cl^{-}
no. of e^{-}s:	18	18	18
no. of protons:	19	16	17

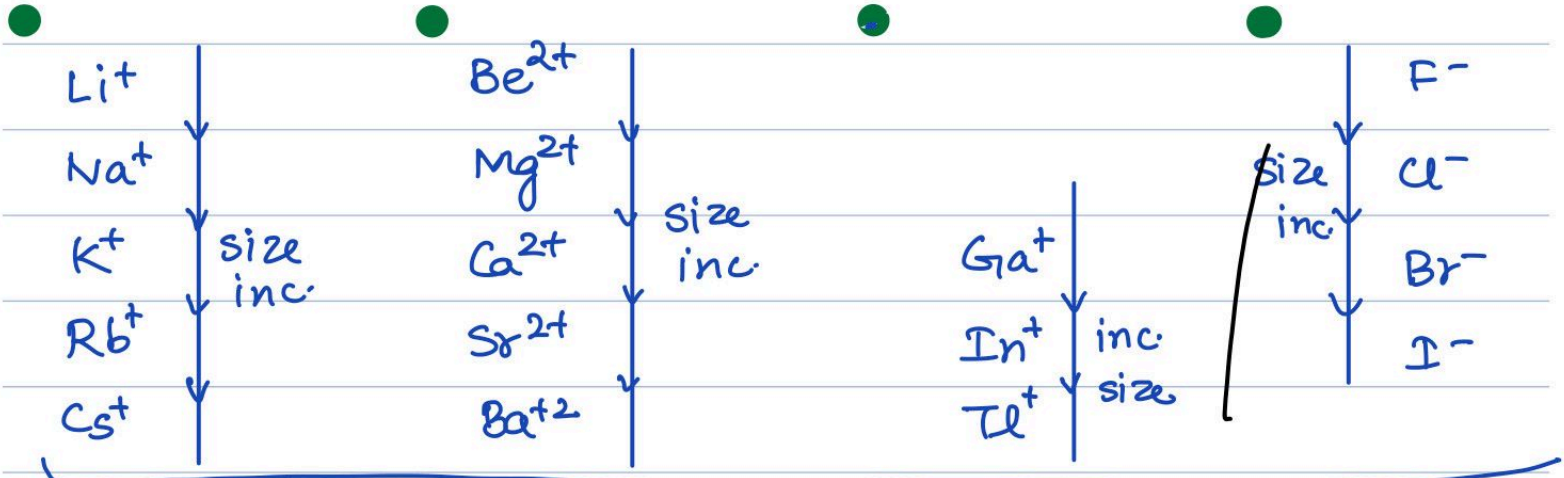
Zeffective: $K^{+} > Cl^{-} > S^{2-}$

Size: $S^{2-} > Cl^{-} > K^{+}$

" v.imp.

For isoelectronic ions, Greater the positive charge, smaller will be size and Greater the negative charge, larger will be Size."

ionic radius comparisons:



In same group, for same charge, size inc. down the group.

- $Li^+ < Be^{2+}$ (Zeff.) | $Li^+ > Be^{2+}$ (size)
- $N^{3-} < O^{2-}$ (Zeff.) | $N^{3-} > O^{2-}$ (size)

v. v. imp. trends of ionic radius to remember:

(i)* $\underbrace{I^-}_{5\text{ shells}} > \underbrace{Br^-}_{1\text{ shell}} > \underbrace{H^-}_{4\text{ shells}} > \underbrace{Cl^-}_{3\text{ shells}} > \underbrace{F^-}_{2\text{ shells}}$

H^- is very large due to low Zeffective. (H^- ← $e^- = 2$, $p = 1$)

(ii)** $\underbrace{Li^+}_{1\text{ shell}} > \underbrace{Mg^{2+}}_{2\text{ shells}} > \underbrace{Al^{3+}}_{2\text{ shells}}$

Li^+ is very large due to low Zeffective

(III). Ionisation Energy:

"Minimum amount of energy required to remove the outermost electron from ground state of an isolated gaseous atom is called ionisation energy of atom."

- Ionisation enthalpy and ionisation potential have exactly same trends as that of ionisation energy.

Reactions releasing energy = exothermic reactions
↳ -ve enthalpy change (ΔH)

Reaction absorbing energy = endothermic reactions
↳ +ve enthalpy change (ΔH)

- For an element,
  |ionisation energy| $\simeq$ |ionisation enthalpy|

Energy is always required to remove electrons

From an atom and hence ionisation enthalpies are always positive

* Ionisation enthalpy is positive for a neutral atom (i.e. endothermic).

* successive ionisation energy values of an element increases continuously (due to dec. size and inc. Z_{eff}) → $z > y > x$

$$Q \xrightarrow{\Delta H = +x} Q^{+} \xrightarrow{\Delta H = +y} Q^{2+} \xrightarrow{\Delta H = +z} Q^{3+}$$

First ionisation energy of Q (IE_1 of Q) = x

First ionisation energy of Q^+ (IE_1 of Q^+) = y

[Second ionisation energy of Q (IE_2)] = y

First ionisation energy of Q^{2+} (IE_1 of Q^{2+}) = z

[Second ionisation energy of Q^+ (IE_2 of Q^+)] = z

[Third ionisation energy of Q (IE_3 of Q)] = z

- Energy required to remove first two electrons from Q = (x+y)

- Energy required to remove first three electrons from Q = (x+y+z)

- Energy req. to remove first two e⁻s from Q^+ = (y+z)

Factors affecting ionisation energy:

(i) **size:** Generally, smaller the size more will be the ionisation energy.

ex: Li $\underset{\text{size}}{>}$ Be

$\underset{IE_1}{<}$

(ii) **Zeffective:** Generally, more Zeffective leads to more ionisation energy.

ex C $\underset{Z_{eff}}{<}$ O

$\underset{IE_1}{<}$

(iii) **stability of electronic configuration:**

- ✓ Exactly half-filled and fully filled subshells provide stability to the configuration.
- ✓ Stable configuration leads to more IE.

[This factor is usually dominating upto adjacent elements]

ex: N: $1s^2\ 2s^2\ 2p^3$ ≷ O: $1s^2\ 2s^2\ 2p^4$ ≷ F: $1s^2\ 2s^2\ 2p^5$



IE_1 order: F > N > O

size: N > O > F
Zeff: N < O < F

(iv) Penetration effect:

Removal of electron from orbital having more penetrating power is more difficult (i.e. more IE)

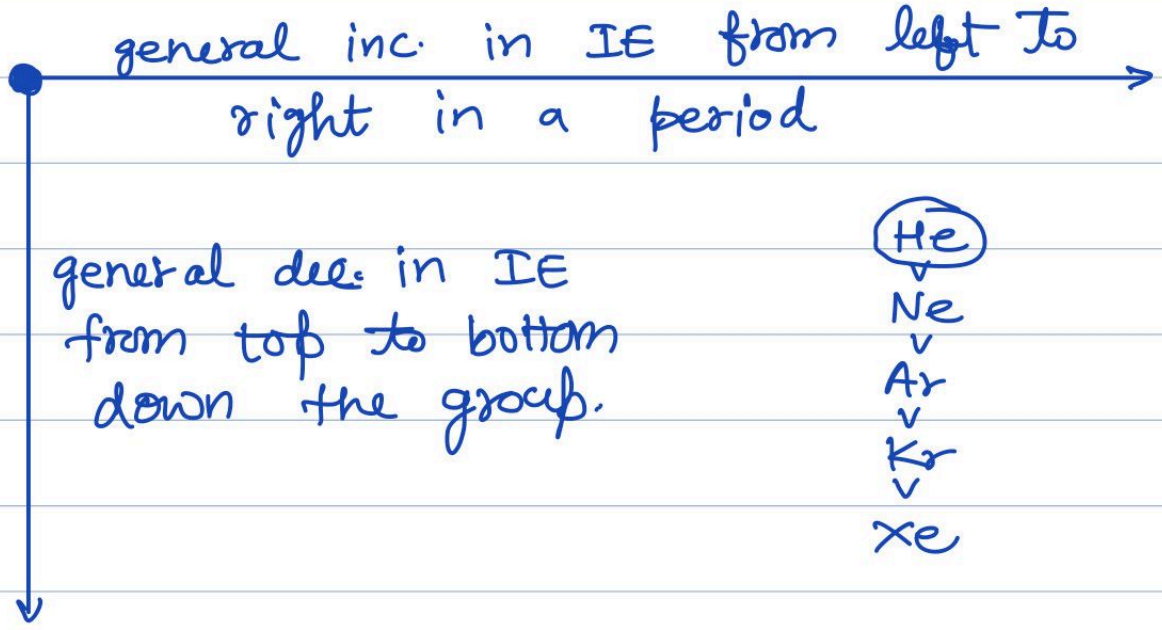
[This factor is usually dominating upto adjacent elements]

ex:

Be $1s^2\ 2s^2$ (↑↓) $>$ IE_1 B $1s^2\ 2s^2\ 2p^1$ (↑ | |)

(∵ 2s is more penetrating than 2p)

General variation of IE in periodic table:



Periodwise variation:

(i). H $1s^1$ $<$ IE He $1s^2$ (stable configuration)

(ii). Li < Be > B < C < N > O < F < Ne

Li: $2s^1$ | Be: $2s^2$ | B: $2p^1$ | C: $2p^2$ | N: $2p^3$ | O: $2p^4$ | F: $2p^5$ | Ne: $2p^6$

Penetration effect (Be, B); stability of configuration (N, O); stability of configuration (F, Ne)

Li < B < Be < C < O < N < F < Ne

(iii). Na < Mg > Al < Si < P > S < Cl < Ar

Na: $3s^1$ | Mg: $3s^2$ | Al: $3p^1$ | Si: $3p^2$ | P: $3p^3$ | S: $3p^4$ | Cl: $3p^5$ | Ar: $3p^6$

Penetration effect (Mg, Al); stability of configuration (P, S); stability of configuration (Cl, Ar)

Na < Al < Mg < Si < S < P < Cl < Ar

IE trends in Gr: 13 and Group: 14:

वो तल गा लि (यों) इंडिया (मे)

B > Tl > Ga > Al > In

Tl: due to high Zeff. (presence of 4f e⁻s)

C > Si > Ge > Pb > Sn

Pb: due to more Zeff. (presence of 4f e⁻s)

v. imp. table:

Maximum IE in a period is for	Noble gas element
Least IE in a period is for	Group-1 element
max IE among all know elements	Helium
Least IE among all know elements	Cs

Ionisation energy in 3d-series:

Sc (21)	Ti (22)	V (23)	Cr (24)	Mn (25)	Fe (26)	Co (27)	Ni (28)	Cu (29)	Zn (30)
$4s^2$ $3d^1$	$4s^2$ $3d^2$	$4s^2$ $3d^3$	$4s^1$ $3d^5$	$4s^2$ $3d^5$	$4s^2$ $3d^6$	$4s^2$ $3d^7$	$4s^2$ $3d^8$	$4s^1$ $3d^{10}$	$4s^2$ $3d^{10}$
$4s^0\,3d^0$	$4s^0\,3d^1$	$4s^0\,3d^2$	$4s^0\,3d^3$	$4s^0\,3d^4$	$4s^0\,3d^5$	$4s^0\,3d^6$	$4s^0\,3d^7$	$4s^0\,3d^8$	$4s^0\,3d^9$

Maximum IE_1 in 3d-series	Zn (stable configuration)
Maximum IE_2 in 3d-series	Cu (e^- to be removed from $3d^{10}$)
Maximum IE_3 in 3d-series	Zn (e^- to be removed from $3d^{10}$)
Maximum IE_4 in 3d-series	Sc (e^- removal from noble gas config.)

Sc^{3+} : [Ar] $4s^0\,3d^0$

IE_2 and IE_3 Comparisons:

- **Compare second ionisation energy in following pairs:**

(a) He (2), Li (3) (b) Be (4), B (5) (c) N (7), O (8) (d) F (9), Cl (17)

Sol.:

(a) He: $1s^2$ | He^+: $1s^1$
Li: $1s^2\,2s^1$ | Li^+: $1s^2$ (more stable)
→ IE_2: Li > He
IE_1: He > Li

(b) Be: $1s^2\,2s^2$ | Be^+: $1s^2\,2s^1$
B: $1s^2\,2s^2\,2p^1$ | B^+: $1s^2\,2s^2$ (more stable)
→ IE_2: B > Be
IE_1: Be > B

(c) N: $1s^2\ 2s^2\ 2p^3$ | N^+: $1s^2\ 2s^2\ 2p^2$

O: $1s^2\ 2s^2\ 2p^4$ | O^+: $1s^2\ 2s^2\ 2p^3$

→ IE_2: O > N

IE_1: N > O

(d)

F: $1s^2\ 2s^2\ 2p^5$ | F^+: $1s^2\ 2s^2\ 2p^4$

Cl: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ | Cl^+: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

→ IE_2: F > Cl

IE_1: F > Cl

Q compare IE_2 & IE_1 for C and N

Sol:

C (6): $1s^2\ 2s^2\ 2p^2$

N (7): $1s^2\ 2s^2\ 2p^3$

IE_1: N > C

C^+: $1s^2\ 2s^2\ 2p^1$

N^+: $1s^2\ 2s^2\ 2p^1$

IE_2: N > C

- **Compare third ionisation energy in following pairs:**

(a) Li (3), Be (4) (b) B (5), C (6) (c) O (8), F (9)

Sol:

(a) Li^{2+} : $1s^1$
Be^{2+} : $1s^2$
IE_3: Be > Li

(b) B^{2+} : $1s^2 2s^1$
C^{2+} : $1s^2 2s^2$
IE_3: C > B

(c) O^{2+} : $1s^2 2s^2 2p^2$
F^{2+} : $1s^2 2s^2 2p^3$
IE_3: F > O

Ionisation energy comparisons between atoms and ions of same element:

[size & Zeffective are dominating over configuration]

(i) $\underset{1s^1}{H} \underset{IE_1}{>} \underset{1s^2}{H^-}$

(ii) $\underset{IE_1}{Cl^+ > Cl > Cl^-}$

IE of cation > IE of atom > IE of anion → for the same element.

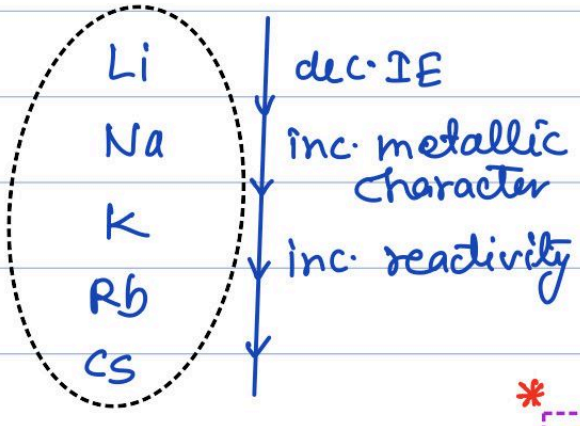
Applications of ionisation energy:

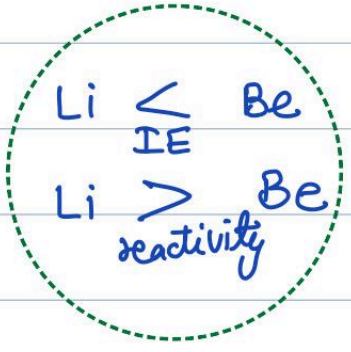
[1]. Metallic character:

Lower the IE → More electropositivity → more metallic character

→ Generally more reactive metal

ex:





Alkali metal < d-block metal (IE)

[2] Prediction of group no. of s and p-block elements on basis of successive IE data:

Maximum difference in	no. of valence e⁻s	Group No.
IE_1 and IE_2	1	1
IE_2 and IE_3	2	2
IE_3 and IE_4	3	13
IE_4 and IE_5	4	14
IE_5 and IE_6	5	15
IE_6 and IE_7	6	16
IE_7 and IE_8	7	17

ex (i) element 'X': IE_1 5.1 eV/atom, IE_2 47 eV/atom, IE_3 69 eV/atom

diff = 47 − 5.1 ≃ 42 eV; diff = 22 eV

Biggest Jump in IE is from IE_1 to IE_2.

No. of valence $e^- = 1$

Group no = 1

ex (ii) element 'Z':

IE_1 8 eV/atom, IE_2 24 eV/atom, IE_3 60 eV/atom, IE_4 250 eV/atom, IE_5 352 eV/atom

Sol

diff = 16 eV; diff = 36 eV; diff 190 eV; diff = 102

No. of valence $e^- = 3$

Group no. = 13

[3]. Stability of oxidation states:

successive IE difference $\geq$ 16 eV	successive IE difference $\leq$ 11 eV	successive IE diff. between 11 To 16 eV
⇓	⇓	⇓
Lower ox. state is more stable	Higher ox. state is more stable	Both ox. states are stable
ex:	ex:	ex:
IE_1 of Na = 5.1 eV	IE_2 of Al = 18 eV	IE_2 of Fe = 16 eV
IE_2 of Na = 47 eV	IE_3 of Al = 28 eV	IE_3 of Fe = 30 eV
Na^{+} > Na^{2+} (stable)	Al^{3+} > Al^{2+} (stable)	Both Fe^{2+} and Fe^{3+} are stable.

[4]. Reducing power:

Generally lower the IE, more is reducing power (provide e^- for reduction of others).

ex

Be
Mg ↓ dec. IE
Ca ↓ inc. reducing power
Sr ↓
Ba

Cs < Na
IE_1
>
Reducing power.

[D]. Electron affinity, Electron gain enthalpy:

Electron affinity (EA)	Electron gain enthalpy ($\Delta_{eg}H$)
The amount of energy released on addition of an electron to the outer most shell of isolated gaseous atom is electron affinity of that atom.	The enthalpy change on addition of an electron to outermost shell of isolated gaseous atom is electron gain enthalpy of that atom.

ex: $A(g) + e^- \longrightarrow A^-(g)$; $\Delta H = -x$ eV

$\Delta_{eg}H$ for $A = -x$

EA for $A \simeq x$

$$\Delta_{eg}H \simeq -EA$$

ex: $Z_{(g)} + e^- \longrightarrow Z^-(g)$, $\Delta H = +y$ eV

$\Delta_{eg}H$ for $Z = +y$

EA for $Z \simeq -y$

ex:

$$\begin{cases} A(g) + e^- \longrightarrow A^-(g); \quad \Delta H = -10, \quad EA \simeq 10 \\ D(g) + e^- \longrightarrow D^-(g); \quad \Delta H = -15, \quad EA \simeq 15 \end{cases}$$

Explanation:

✓ Electron affinity of D is more than that of A.

✓ D has more -ve electron gain enthalpy than that of A.

$$X(g) \xrightarrow[\Delta H_1]{+e^-} X^{\ominus}(g) \xrightarrow[\Delta H_2]{+e^-} X^{2-}(g)$$

ΔH_1 → $\Delta_{eg_1}H$ of X

ΔH_2 → $\Delta_{eg_2}H$ of $X^{\ominus}$, $\Delta_{eg_2}H$ of X

very important points:

*** [i]. $\Delta_{eg_1}H$ for s and p-block elements

positive for

- ✓ Noble gas elements (He, Ne, Ar, Kr, Xe, ...) ($ns^2 np^6$)
- ✓ Group-2 metals (Be, Mg, Ca, Sr, Ba--) (ns^2)
- ✓ Nitrogen element ($2p^3$) (+ more interelectronic repulsion)

Negative for

- ✓ Rest elements of s and p-block

**

[ii]. Addition of an electron to a cation is "Exothermic" ∵[attraction b/w e^- being added & ion]

Δeg_1H of a cation = -ve

**

[iii]. Addition of an electron to an anion is "Endothermic" ∵[Repulsion b/w e^- being added & ion]

Δeg_1H of an anion = +ve

(∵ Δeg_2H, Δeg_3H, ... are positive for any atom)

****

[iv]. Addition of two or more electrons to an atom is always endothermic.

ex:

$$X(g) + e^- \longrightarrow X^{\ominus}(g), \quad \Delta H_1 = -P_1$$

$$X^{\ominus}(g) + e^- \longrightarrow X^{2-}(g), \quad \Delta H_2 = +P_2$$

$$X(g) + 2e^- \longrightarrow X^{2-}(g), \quad \Delta H_T = (P_2 - P_1)$$

↓

+ve (∵ $|P_2| > |P_1|$)

ex:

$$O(g) + 2e^- \longrightarrow O^{2-}(g), \quad \Delta H = +ve$$

$$N(g) + 3e^- \longrightarrow N^{3-}(g), \quad \Delta H = +ve$$

Fill the following table:

Species	Process	Sign of ΔH
Atom	Removal of 1 or more electrons	+ve
Atom	Addition of 1 electron	+ve or -ve
Atom	Addition of 2 or more electrons	+ve
Cation	Removal of 1 or more electrons	+ve
Cation	Addition of 1 electron	-ve
Cation	Addition of 2 or more electrons	+ve or -ve
Anion	Addition of 1 or more electrons	+ve
Anion	Removal of 1 or more electrons	+ve

Factors affecting electron affinity:

(i). **Size:** smaller the size of atom, more attractive force on coming electron, generally more energy is released.

(ii). **Effective nuclear charge:** more Zeffective, generally more energy released.

(iii). **Stability of configuration:** exactly half-filled or fully filled configuration ⟶ less tendency to accept e^- [either +ve $\Delta_{eg}H$ (or) -ve $\Delta_{eg}H$, less magnitude

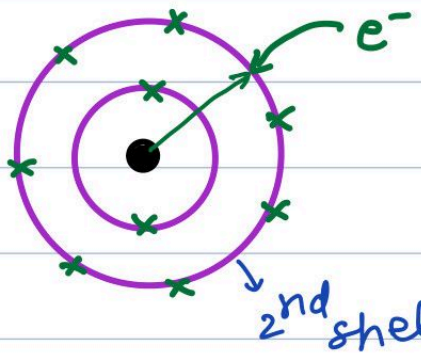
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(iv). **Interelectronic repulsion:**

More interelectronic repulsion may lead to lower electron affinity.

ex:

F(9) : 2,7



more e^- density in the shell to which electron is being added

↓

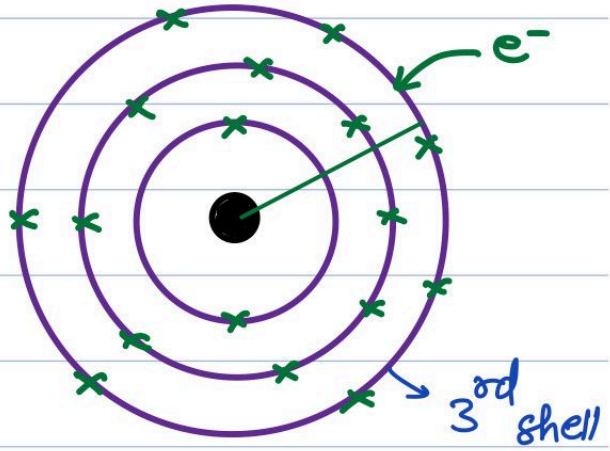
more repulsion to coming electrons by already present electrons

↓

Less energy release

[Less EA]

Cl(17) : 2, 8, 7



Less e^- density in the shell to which e^- is being added,

↳ less repulsion to coming e^-.

↓

[More EA]

EA in p-block :

2nd Period	B	C	N	O	F
3rd Period	Al	Si	P	S	Cl

Due to more interelectronic repulsions, the second period elements have lower EA than corresponding third period elements in p-block.

Electron affinity orders to remember :

(i) Group No: 17 : $Cl > F > Br > I$

(ii) Group No: 16 : $S > Se > Te > Po > O$

EA comparison in Periods:

(i) $H > He$

(ii) $Li > \underset{2s^2}{Be} < B < C > \underset{2p^3}{N} < O < F > \underset{2p^6}{Ne}$

(iii) $Na > \underset{3s^2}{Mg} < Al < Si > \underset{3p^3}{P} < S < Cl > \underset{3p^6}{Ar}$

- In respective periods, generally halogens have maximum electron affinity. (Reason: high Z_{eff}, smaller size)

- "Cl has maximum electron affinity among all known elements."

- For same period,

[Group-1 metal (ns^1) $\underset{EA}{>}$ Group-13 element (np^1)]

ex: $\begin{cases} Li \underset{EA}{>} B \\ Na \underset{EA}{>} Al \end{cases}$

some v. imp points :

(i) $|\Delta_{eg_1}H \text{ of atom}| = |\Delta_{ionisation}H \text{ of anion}|$

→ for same element & its anion.

ex:

$Cl + e^- \longrightarrow Cl^-$, $\Delta H = -x$; [e^- gain of Cl atom

$Cl^- \longrightarrow Cl + e^-$, $\Delta H = +x$; [ionisation of Cl^- ion

(ii) $|\Delta_{ionisation}H \text{ of atom}| = |\Delta_{eg_1}H \text{ of cation}|$

→ For same element & its cation

ex:

$Na \longrightarrow Na^+ + e^-$, $\Delta H = +y$ [ionisation of Na atom

$Na^+ + e^- \longrightarrow Na$, $\Delta H = -y$ [e^- gain of Na^+ ion

(iii) IE_1 of F^- < IE_1 of Cl^-

Similarly, [IE_1 of O^- < IE_1 of S^-]

explanation:

$F + e^- \longrightarrow F^-$, $\Delta H = -x$

$Cl + e^- \longrightarrow Cl^-$, $\Delta H = -y$

$|y| > |x|$

$F^- \longrightarrow F + e^-$, $\Delta H = +x$

$Cl^- \longrightarrow Cl + e^-$, $\Delta H = +y$

IE_1 of Cl^- > IE_1 of F^-

IE_1 : $F > Cl > Cl^- > F^-$

[while comparing IE_1 of anions from same group, use reverse of Electron affinity]

◎ while comparing EA of cations of same group, use Reverse of IE.

ex: $F \longrightarrow F^+ + e^-$, (more absorb) $\left\{ F^+ + e^- \longrightarrow F \text{ (more release)} \right.$

$Cl \longrightarrow Cl^+ + e^-$, $\left\{ Cl^+ + e^- \longrightarrow Cl \right.$

$[F^+ \underset{EA}{>} Cl^+]$

Species	Process	Sign of ΔH
Atom	Removal of 1 or more electrons	+ve
Atom	Addition of 1 electron	+ve (or) -ve
Atom	Addition of 2 or more electrons	+ve
Cation	Removal of 1 or more electrons	+ve
Cation	Addition of 1 electron	-ve
Cation	Addition of 2 or more electrons	+ve (or) -ve
Anion	Addition of 1 or more electrons	+ve
Anion	Removal of 1 or more electrons	+ve (or) -ve

Q. $O \longrightarrow O^{2-}$ (iv) $N \longrightarrow N^{3-}$ is endothermic. Still oxides (O^{2-}) (or) nitrides (N^{3-}) are abundant in nature. Why?

Sol:

$O \xrightarrow[\text{endo}]{} O^{2-}$ — compensates for energy given via energy released on bond formation.

$N \xrightarrow[\text{endo}]{} N^{-3}$ — compensates for energy given via energy released on bond formation.

Q. Which of the following ion is the least stable?

(A) Li^- (B) C^- (C) Be^- (D) F^-

Sol: ANS: (C)

$$\begin{cases} Li + e^- \longrightarrow Li^- \ (exo) \\ C + e^- \longrightarrow C^- \ (exo) \\ F + e^- \longrightarrow F^- \ (exo) \end{cases}$$

$$\{ Be + e^- \longrightarrow Be^- \ (endo)$$

Multiple correct

Q. Which of the following processes are exothermic

(A) $O^{2-}(g) \longrightarrow O(g)$ (B) $Be^-(g) \longrightarrow Be(g)$

(C) $P(g) \longrightarrow P^{-3}(g)$ (D) $H^+(g) \longrightarrow H^-(g)$

Sol: [ANS: (A, B, D)]

(A) $O \longrightarrow O^{2-}$ (endo) $\}$ $O^{2-} \longrightarrow O$ (exo)

$Be \longrightarrow Be^-$ (endo) $\}$ $Be^- \longrightarrow Be$ (exo)

$P \longrightarrow P^{-3}$ endo

$H^+ + e^- \longrightarrow H$ (exo)

$H + e^- \longrightarrow H^-$ (exo)

$H^+ + 2e^- \longrightarrow H^-$ (exo)

[E]. Electronegativity (χ):

The tendency of a covalently bonded atom to attract the shared electron pairs towards itself is electronegativity.

- X—Z
  - ↳ shared $\bar{e}$ pair at about middle position

  } Electronegativity of X is equal electronegativity of Z.

- $\delta-$ J — Q $\delta+$
  - ↳ shared $\bar{e}$ pair is polarised more towards J.

  } Electronegativity of J is more than that of Q.
  [Due to this electronegativity difference, polarity is arised in the bond]

Characteristics of electronegativity:

(i). It is property of bonded atom not of isolated atom.

(ii). It is unitless, dimensionless.

(iii). It is a variable property, relative property.

Scales of electronegativity: (only for JEE-advanced, KVPY, olympiad)

(i) Most followed

Pauling's scale:

✓ based on bond energy data

Let:

$\delta+$ A —— B $\delta-$

Resonace energy (Δ) = Experimental bond dissociation energy (E_{A-B}) − Theoretical bond dissociation energy $\sqrt{E_{A-A} \times E_{B-B}}$

Geometric mean

✓✓ $|\chi_B - \chi_A| = \sqrt{\Delta}$ (Δ in ev/atom)

Reference:

Electronegativity of F = 4.0 [acc. to NCERT]

1 ev/atom = 96.46 KJ/mol

[1 Cal = 4.18 J]

EN values on Pauling's scale:

H 2.1						
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
		Al 1.5	Si 1.8	P 2.5	S 2.8	Cl 3.0
						Br 2.8
						I 2.5

* F > O > N ≃ Cl > C ≃ I > ···· > Cs

(ii) Mulliken's scale:

$$\chi_M = \frac{(IE + EA)}{2}$$

→ IE and EA in eV/atom

For an element:

$$\chi_M \simeq 2.8\,\chi_P$$

χ_M: Mulliken's EN
χ_P: Pauling's EN

$$\chi_P = \frac{(IE + EA)}{2 \times 2.8}$$

→ IE & EA in eV/atom

[iii]. Alred-Rochow scale:

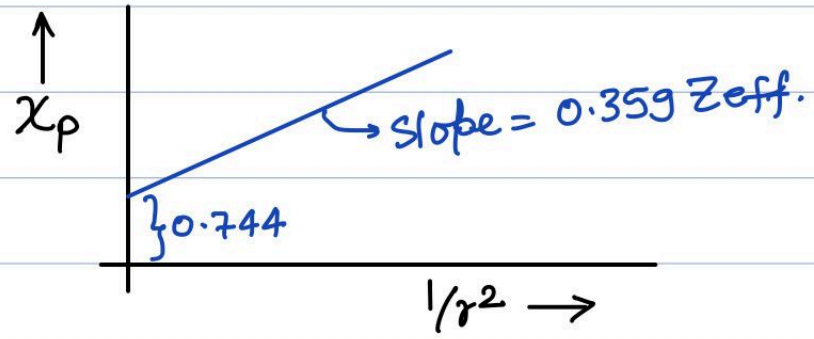
$$\chi_{AR} = \frac{0.359\,Z_{eff.}}{r^2}$$

Z_{eff} = effective nuclear charge
r = covalent radius (Å)

For an element,

$$\chi_P \simeq 0.744 + \chi_{AR}$$

$$\chi_P \simeq 0.744 + 0.359\,\frac{Z_{eff}}{r^2}$$



Factors affecting electronegativity:

(i). **Size:** smaller the size, generally more EN.

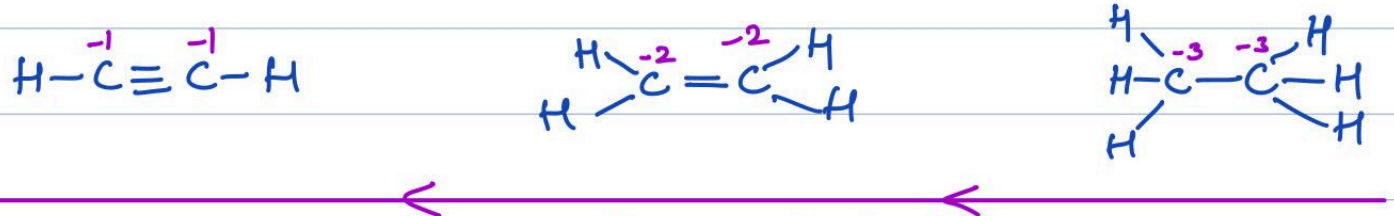
(ii). **Zeffective:** More Zeffective, generally more EN.

(iii). **Oxidation no.** For same element,

- more +ve ox. no. ⟹ more EN
- more -ve ox. no. ⟹ less EN

(iv). **Bond multiplicity (% s-character):** → in next chapter

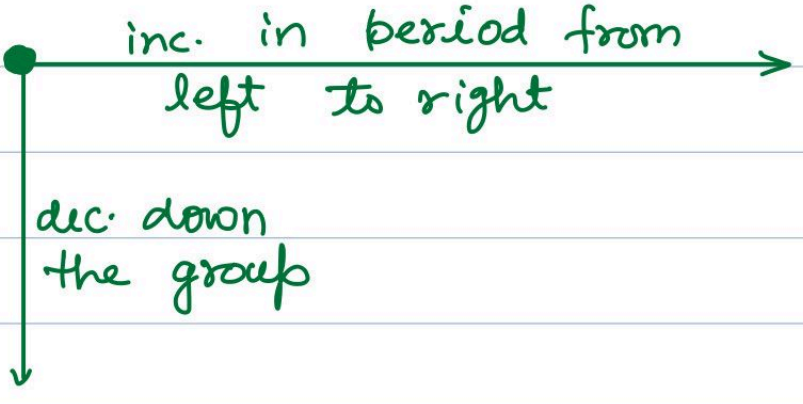
for same element, more bond multiplicity, more EN.



inc. bond multiplicity

inc. EN of C

General trends in EN:



2nd period:

Li < Be < B < C < N < O < F

3rd period:

Na < Mg < Al < Si < P < S < Cl

In any period:

(≠ Period=1)

- most EN → Halogen
- Least EN → Alkali metal

EN order to remember:

F > O > N ≃ Cl > C ≃ S ≃ I > P ≃ H > ----- > Cs

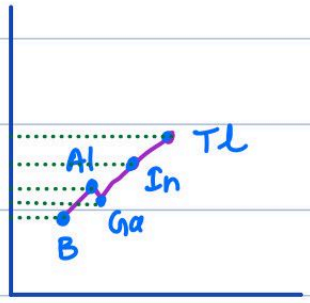
S > C (slightly)

P > H (slightly)

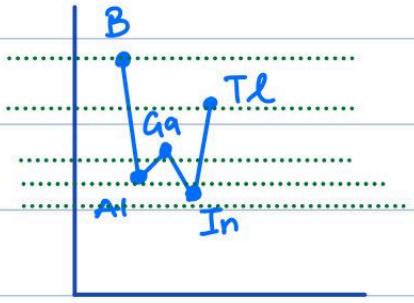
p- Block Groups which do not follow general variation of EN: (not so imp.)

Gr: 13	Gr: 14
B (2.0)	C (2.5)
Al (1.5)	Si (1.8)
Ga (1.6)	Ge (1.8)
In (1.7)	Sn (1.8)
Tl (1.8)	Pb (1.8)

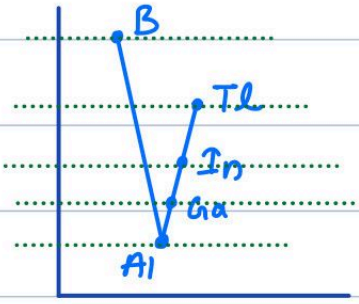
Size



$|E_1|$



EN



Applications of EN:

(i) % ionic character in a covalent bond:

$$\overset{\delta+}{A}\text{——}\overset{\delta-}{B}$$

{ more EN difference b/w A and B, more partial charge developed on them ⟹ more ionic character in bond.

ex: $\overset{\delta+}{H}\text{—}\overset{\delta-}{F}$ $\quad$ $\overset{\delta+}{H}\text{—}\overset{\delta-}{Cl}$ $\quad$ $\overset{\delta+}{H}\text{—}\overset{\delta-}{Br}$

$\longleftarrow$ inc. EN difference

inc. % ionic character in bond

(ii) Metallic and non-metallic nature:

{ EN : directly related to non-metallic character

{ EN : inversely related to metallic character

Q. Compare non-metallic nature of K, Si, Cl

Sol: $K < Si < Cl$ (as per EN trend)

Nature of oxides:

[A]. Acidic oxides: oxides which react with bases to form salt are acidic oxides.

* Most of the non-metal oxides are acidic.

ex: CO_2, SO_2, SO_3, Cl_2O_7, B_2O_3, SiO_2, NO_2, N_2O_5, ...

Oxides of metals or metalloids in which ox. no. of metal or metalloid is greater or equal to +5 are acidic.

ex: $\overset{+6}{MnO_3}$, Mn_2O_7 ($\to +7$ each), $\overset{+6}{CrO_3}$, Bi_2O_5 ($\to +5$ each), As_2O_5 ($\to +5$ each), Sb_2O_5 ($\to +5$ each) ...

(ox. no. of each oxygen in oxide = -2)

comparison of acidic strength in oxides:

[more EN $\longrightarrow$ more non-metallic nature $\longrightarrow$ more acidic]

(i) $\overset{+4}{SO_2} < \overset{+6}{SO_3}$

(ii) $SO_2 > CO_2$ (S slightly more EN than C)

(iii) $\overset{+5}{N_2O_5} > \overset{+4}{N_2O_4} > \overset{+3}{N_2O_3}$

(iv) $P_4O_{10} < SO_3 < Cl_2O_7$ (EN: P<S<Cl)

(v) $CO_2 < N_2O_5$ (EN: N>C)

[B]. Basic oxides: oxides which react with acids to form salt are basic oxides

- Metal oxides in which ox. no. of metal is less than +5 are generally basic.

ex: $\overset{+1}{Na_2O}$, $\overset{+2}{CaO}$, $\overset{+3}{Fe_2O_3}$, $\overset{+2}{CuO}$, $\overset{+1}{Cu_2O}$, $\overset{+2}{HgO}$, $\overset{+2}{MnO}$, $\overset{+2}{CrO}$, $\overset{+2}{FeO}$

comparison of basic strength in oxides:

[less EN ⟶ more metallic nature ⟶ more basic]

(i) $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$

(ii) $Na_2O > MgO$

(iii) $Na_2O > CuO$

(iv) $\overset{+2}{FeO} > \overset{+3}{Fe_2O_3}$

[C]. Amphoteric oxides:

oxides which react with both acids and bases are amphoteric oxides.

List of amphoteric oxides:

s-block

BeO

p-block

Gr: 13	Gr 14	Gr 15
Al_2O_3	SnO, SnO_2	As_2O_3 (+3)
Ga_2O_3	PbO, PbO_2	Sb_2O_3 (+3)
	Pb_2O_3, Pb_3O_4	
	(all oxides of Sn & Pb)	

d-block:

ZnO, Cr_2O_3 (+3), MnO_2 (+4), V_2O_5 (+5), TiO_2 (+4),

Respective Hydroxides of above shown oxides are also amphoteric.

[D]. Neutral oxides:

Oxides which do not react with acids or bases to form salt are neutral oxides.

Neutral oxides:

- NO (Nitric oxide)
- N_2O (Nitrous oxide)
- CO (carbon monoxide)
- H_2O

Diagonal relationship: few elements of second period resembles with diagonally placed third period elements. This is called diagonal relationship.

	Gr-1	Gr-2	Gr-13	Gr-14
2nd Period:	Li	Be	B	C
3rd Period:	Na	Mg	Al	Si